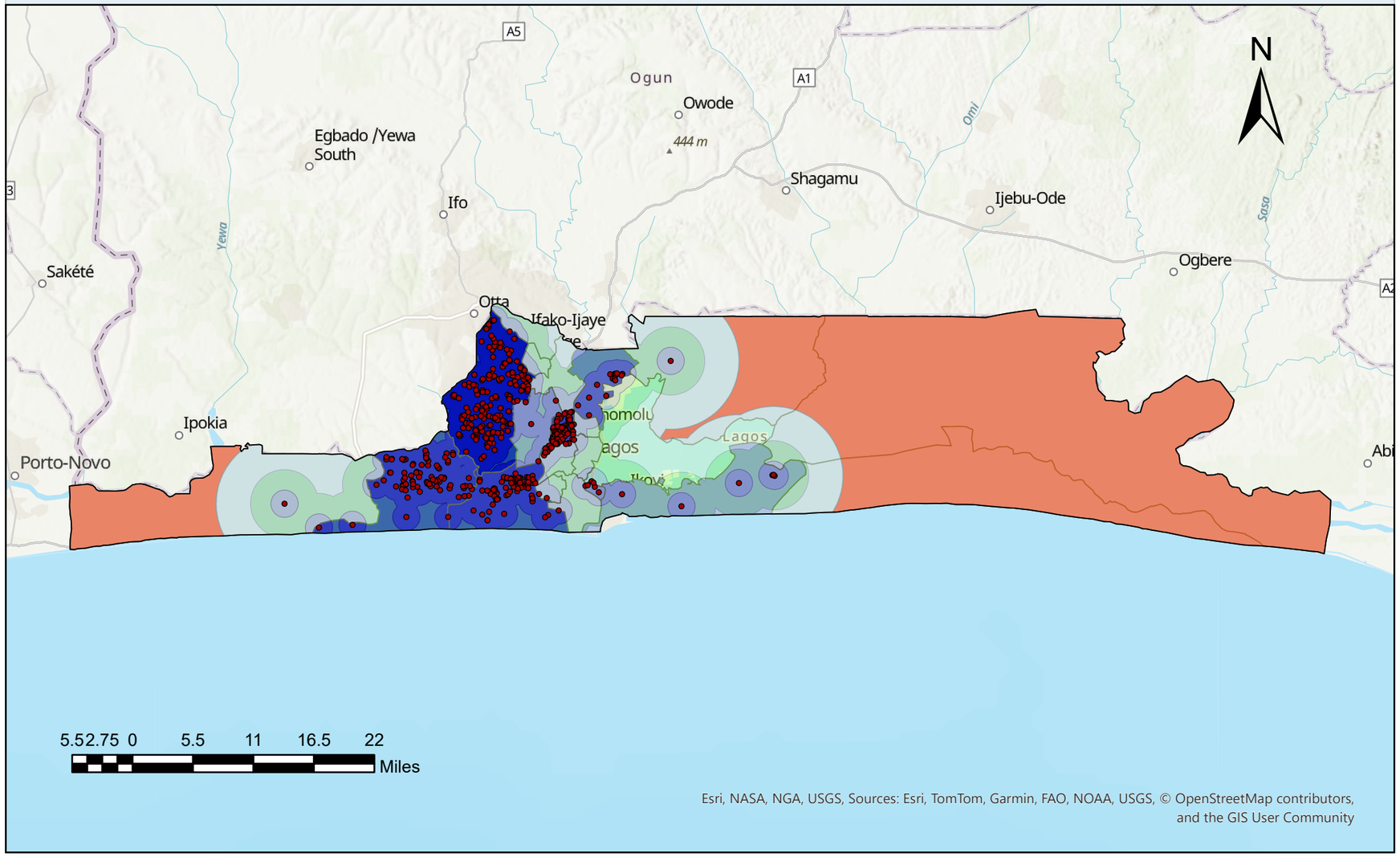
EMERGENCY HEALTHCARE ACCESSIBILITY & SERVICE COVERAGE ANALYSIS

Lagos State, Nigeria | GIS-Based Assessment of Healthcare Facility Distribution and Spatial Accessibility

MAIN MAP — HEALTHCARE ACCESSIBILITY, FACILITY DISTRIBUTION & SERVICE COVERAGE



KEY RESULTS

493
Mapped hospital/clinic points

1,978.35 km²
Area >10 km from mapped facility

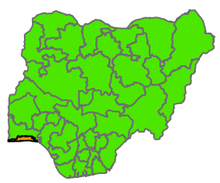
52.61%
Potentially underserved share

180
Highest LGA count — Alimosho

LEGEND

- Potentially underserved (>10 km)
- 0–2 km | Higher accessibility
- 2–5 km | Moderate accessibility
- 5–10 km | Lower accessibility
- Mapped hospital / clinic
- LGA facility count (graduated blue)

LOCATOR MAP



METHODOLOGY

OpenStreetMap/HOT healthcare data were filtered to hospital and clinic features, clipped to Lagos State and checked for duplicate point/polygon representations. A validated set of 493 mapped facilities was retained. Multiple-ring buffers of 2, 5 and 10 km were created as straight-line accessibility screening zones and clipped to the study area.

Areas beyond 10 km were mapped as potentially underserved. ADM2 boundaries were spatially joined with facility points to compare mapped facility counts across Lagos LGAs.

DATA SOURCES

- OpenStreetMap / HOT Export — hospitals & clinics
- geoBoundaries — Nigeria ADM2 / Lagos boundary
- World Topographic Map / World Hillshade

Software: ArcGIS Pro

LIMITATIONS & MAP INFORMATION

Screening assessment only — not a travel-time, capacity or healthcare-quality model. Distances are straight-line and do not account for road travel time, congestion, service capacity, opening hours or population demand. Results dependon completeness and positional accuracy of mapped OpenStreetMap facilities. A zero LGA count means no mapped pointwas identified in the source dataset, not that no healthcare facility exists.

CRS: WGS 1984 UTM Zone 31N | Lagos State study area: 3,760.19 km² | November 2026